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NEET PYQ Based Test (Electric Charges & Fields)
50 Questions — NEET Level + Rank Booster

Section A: MCQs (1–30)

1. Coulomb's law is valid for: (a) point charges (b) extended bodies (c) both (d) none
2. SI unit of electric charge: (a) Volt (b) Coulomb (c) Ampere (d) Ohm
3. Charge on electron: (a) $+1.6 \times 10^{-19}C$ (b) $-1.6 \times 10^{-19}C$ (c) 0 (d) $1C$
4. Force between two charges depends on: (a) product of charges (b) distance (c) medium (d) all
5. Electric charge is: (a) scalar (b) vector (c) tensor (d) none
6. In a conductor, charges reside: (a) inside (b) surface (c) both (d) none
7. Earthing is used to: (a) remove charge (b) add charge (c) neutralize (d) both a & c
8. Insulators: (a) allow free flow (b) resist flow (c) partially allow (d) none
9. Unit of permittivity: (a) C^2/Nm^2 (b) Nm^2/C^2 (c) N/C (d) none
10. Coulomb constant k equals: (a) 9×10^9 (b) 8.85×10^{-12} (c) 10^6 (d) none
11. Two charges doubled, force becomes: (a) same (b) double (c) four times (d) half
12. Distance doubled, force becomes: (a) $F/2$ (b) $F/4$ (c) $2F$ (d) $4F$
13. Charge quantization is: (a) continuous (b) discrete (c) both (d) none
14. Gold leaf electroscope detects: (a) current (b) charge (c) voltage (d) resistance
15. Method of charging without contact: (a) friction (b) conduction (c) induction (d) heating
16. Neutral body has: (a) zero charge (b) equal + and - (c) no electrons (d) none

Section B: Assertion–Reason (31–40)

Directions: (A) Both correct, R explains (B) Both correct, not explanation (C) A correct, R wrong (D) A wrong, R correct

31. Assertion: Electric charge is conserved.
    Reason: Total charge remains constant.
32. Assertion: Coulomb law follows inverse square.
    Reason: Force $\propto r$
33. Assertion: Insulators do not conduct.
    Reason: No free electrons.
34. Assertion: Induction requires contact.
    Reason: Charges transfer directly.
35. Assertion: Force depends on medium.
    Reason: Permittivity changes.

Section C: Case Study (41–45)

Case: Two charges $+q$ and $-q$ are placed at distance d.

41. Nature of force: (a) repulsive (b) attractive (c) zero (d) none
42. Force magnitude: (a) kq^2/d^2 (b) kq/d (c) kq^2d (d) none
43. If distance doubled: (a) $F/4$ (b) $F/2$ (c) $2F$ (d) $4F$
44. If medium changes: (a) force same (b) decreases (c) increases (d) zero
45. Net field at midpoint: (a) zero (b) maximum (c) infinite (d) none

Section D: Rank Booster Numericals (46–50)

46. Two charges $+2\mu C$ and $+8\mu C$ are 1 m apart. Find point where net field zero.
47. Two electrons at $10^{-10}m$. Find force.
48. Sphere charge Q, split into two. Find max force condition.
49. Charge $5 \times 10^{-9}C$. Find number of electrons.
50. Force becomes when distance becomes $r/3$.